**Department of Electronics and Electrical Communication Engineering**
Indian Institute of Technology Kharagpur

# Digital Communication Laboratory

## EC39002

Instructors: Prof. Ritwik Kumar Layek, Prof. Amitalok Budkuley, and Prof. Jithin Ravi

## Experiment 3: Binary Phase Shift Keying (BPSK) Modulation and Demodulation




**Jai Raj**
23EC10031

Group 9

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# 1 Objectives

The objectives of this experiment are to:

1. Implement Binary Phase Shift Keying (BPSK) modulation for a random bit sequence.
2. Design and implement a demodulator to recover the original binary sequence from the modulated signal.
3. Observe and analyze the time-domain waveforms of the modulator and demodulator using an oscilloscope.

# 2 Key Components

- **Integrated Circuits:**
  - IC 741 Op-Amp (used in the modulation/demodulation stages)
  - IC M4016 Analog Switch
  - IC 7486 (Quad 2-input XOR gate)
  - IC 7495 (4-bit Shift Register) for PN sequence generation
- **Passive Components:** Resistors (various values for gain and switching logic).
- **Equipment:** Mixed-signal oscilloscope (Tektronix 2 Series), function generator.

# 3 Brief Theory

## 3.1 Phase Shift Keying (PSK)

Phase Shift Keying (PSK) is a digital modulation scheme that conveys data by changing (modulating) the phase of a constant-frequency reference signal (the carrier). It is widely used in modern communication standards such as RFID, Bluetooth, and wireless LANs.

## 3.2 Binary Phase Shift Keying (BPSK)

In BPSK, the phase of the carrier signal is shifted between two states: $0^\circ$ and $180^\circ$ ($\pi$ radians).

- **Binary '1':** Represented by the carrier signal with $0^\circ$ phase shift.
- **Binary '0':** Represented by the carrier signal with $180^\circ$ phase shift (phase reversal).

Mathematically, the BPSK signal $s(t)$ can be represented using a single basis function:

$$s(t) = m(t)\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t) \tag{1}$$

where $m(t)$ is the polar NRZ message signal taking values $+1$ or $-1$. Inverting a sinusoid is equivalent to a 180° phase shift; thus, a BPSK modulator can be thought of as a balanced modulator (multiplier) that multiplies the carrier with the binary data.

# 4 Modulator and Demodulator Design

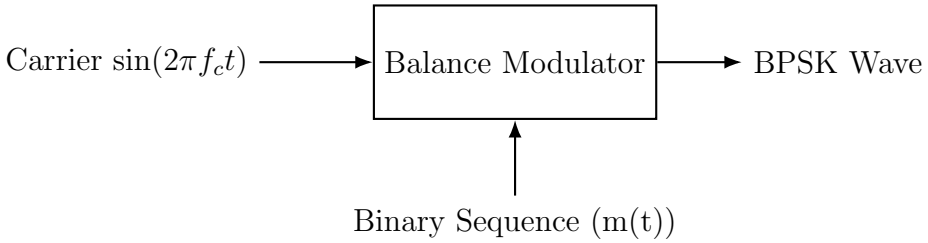



Figure 1: Conceptual Block Diagram of BPSK Modulator

The modulation is achieved using an analog switch (IC M4016) or a balanced modulator that selects between the original carrier and an inverted version of the carrier based on the input bit.

## 4.1 Circuit Design: Modulator

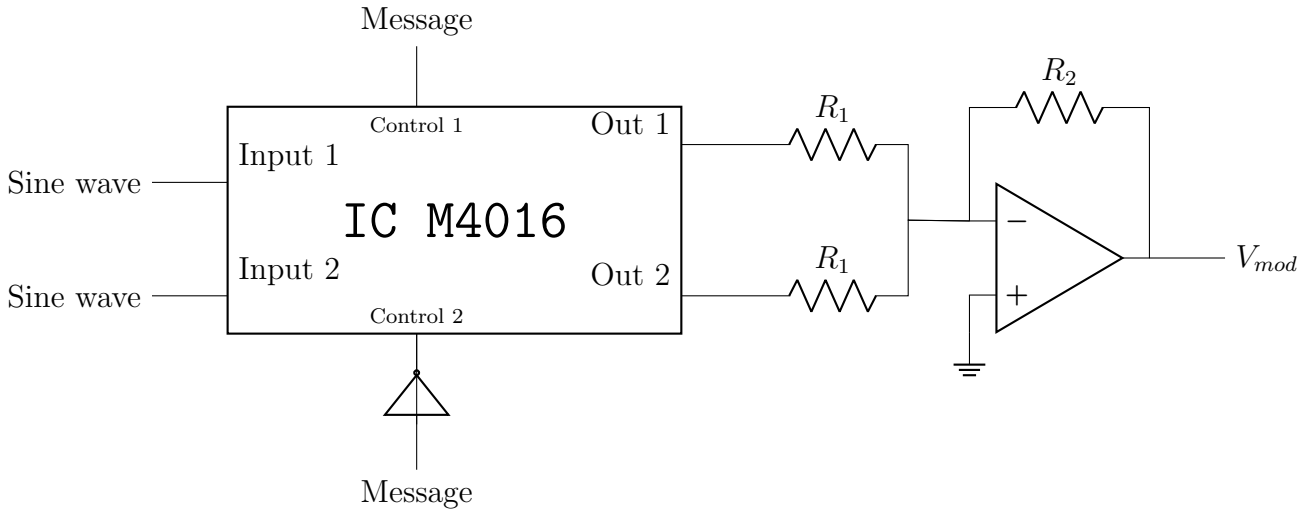



Figure 2: BPSK Modulator Circuit

## 4.2 Circuit Design: Demodulator

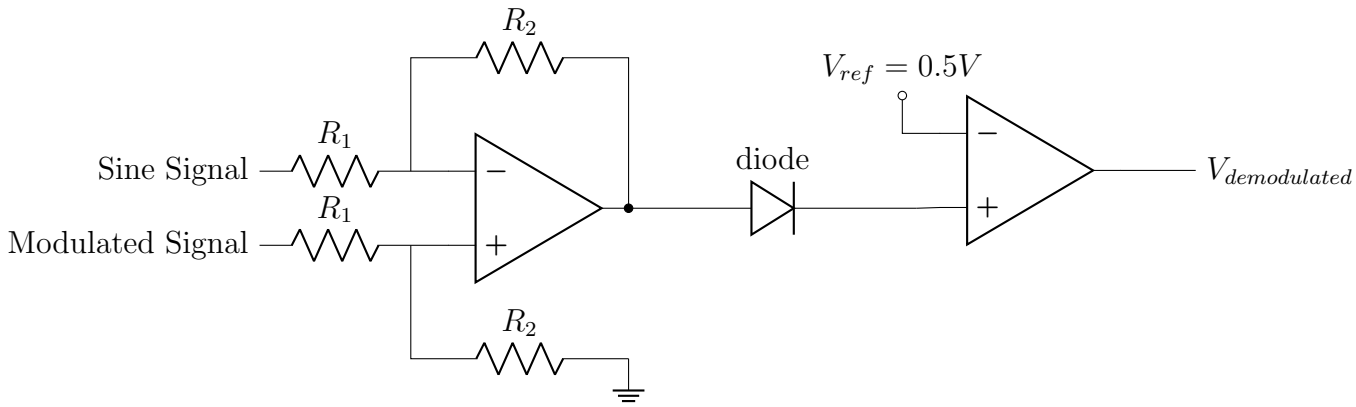



Figure 3: BPSK Demodulator Circuit

# 5 Observations and Results

## 5.1 BPSK Modulation Waveforms

The following oscilloscope traces show the input message sequence (Yellow) and the resulting BPSK modulated output (Blue). Phase reversals are clearly visible at the bit transitions.




Figure 4: BPSK modulation waveform (Yellow: message signal, Blue: BPSK modulated signal). Here the carrier sine-wave frequency is an integer multiple of the message signal frequency.

Figure 4 shows that the blue modulated waveform follows the yellow message sequence: within each bit interval the carrier remains sinusoidal, and a 180° phase reversal occurs when the message bit changes state.

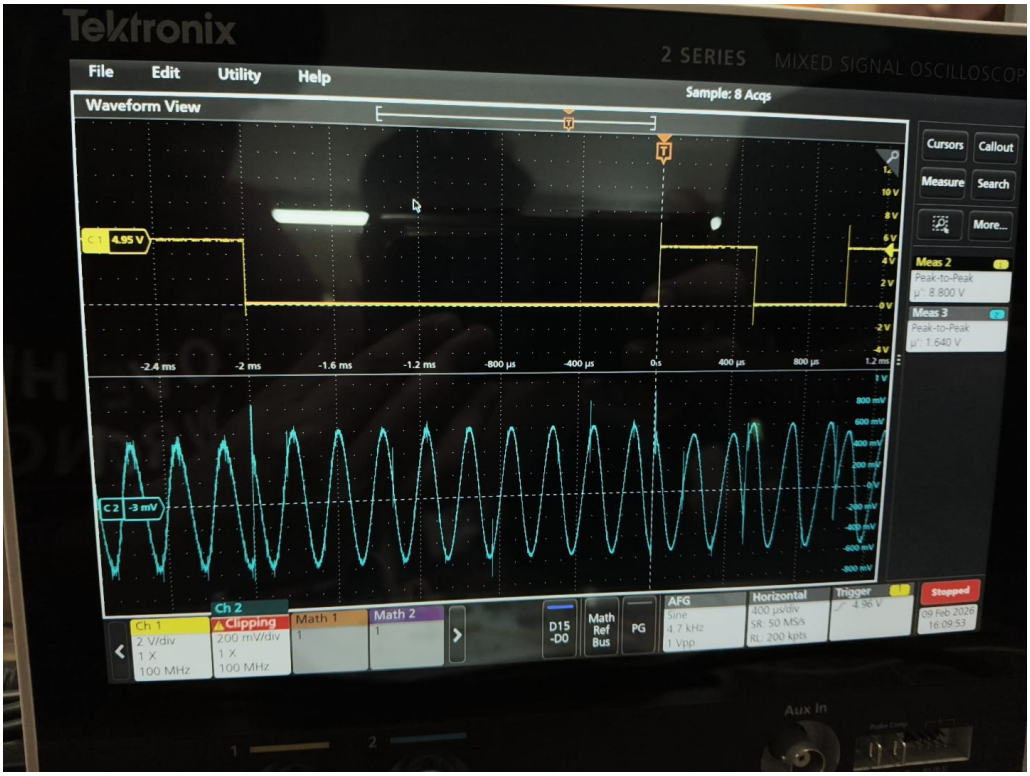



Figure 5: BPSK modulation waveform (Yellow: message signal, Blue: BPSK modulated signal) when the carrier sine-wave frequency is a non-integer multiple of the message signal frequency.

In Figure 5, the phase inversion is still visible at bit transitions, but the carrier does not complete an integer number of cycles per bit interval due to the non-integer frequency ratio.

## 5.2 BPSK Demodulation and Data Recovery

The demodulator circuit was implemented to recover the original message. As shown in the oscilloscope capture, the recovered output (Yellow) matches the input PN sequence.

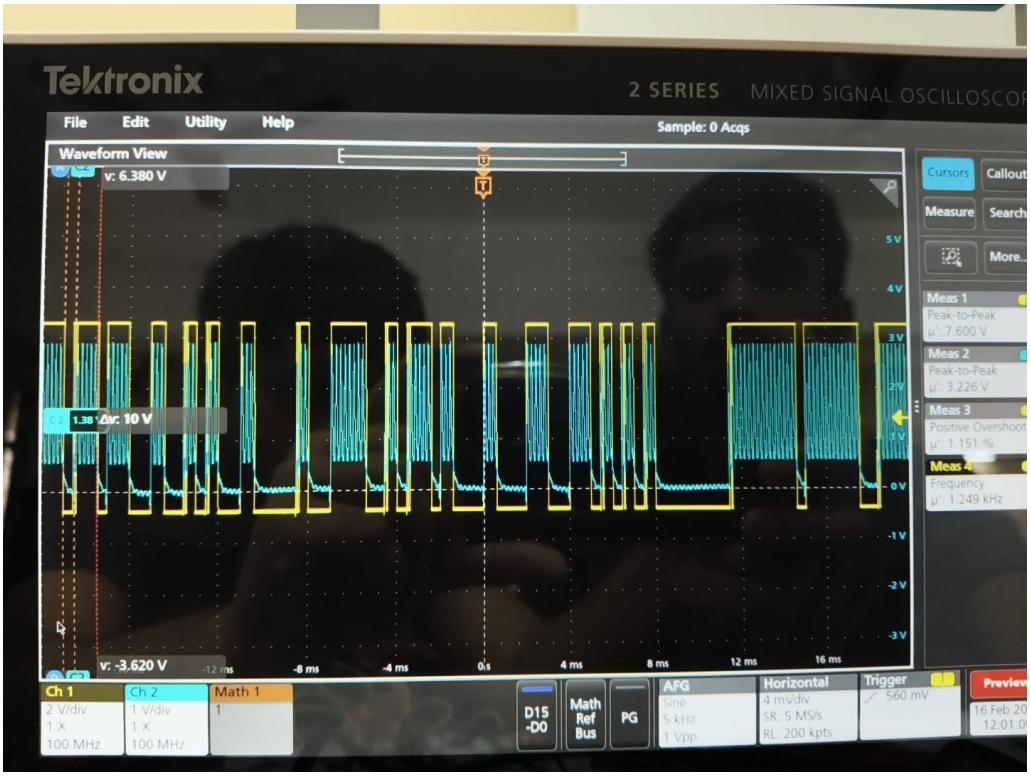



Figure 6: Oscilloscope Trace: Recovered Binary Sequence from Demodulator

# 6 Discussion

- **Signal sources used:** The PN (message) sequence was generated using the signal generator, and the sine (carrier) signal was obtained from the function generator/oscilloscope.
- **Modulator implementation (IC M4016):** The BPSK modulator was implemented using the IC M4016 analog switch. The control signals for the two switches were the message and its inverted version. Both switch inputs were fed with the same carrier sine wave (no phase shift), and the two switch outputs were added using an op-amp adder (gain $> 1$) to obtain the desired BPSK modulated signal.
- **Effect of carrier-to-message frequency ratio:** When the sine-wave frequency was an integer multiple of the message signal frequency, the carrier completed an integer number of cycles within each bit interval before inverting. For the non-integer multiple case, the carrier inverted at irregular points and did not complete an integer number of cycles per bit interval.
- **Simplification observed:** We later realized that the op-amp adder stage is not strictly necessary: since the inactive M4016 switch output is effectively high-impedance (Hi-Z), the two switch outputs can be tied together directly without shorting.
- **Demodulation approach:** The received signal was first demodulated by subtracting the carrier sine wave from the modulated waveform. This produced an output close to zero when both were in phase and approximately a $2\times$ sine wave when they were $180^\circ$ out of phase.

- **Diode and comparator thresholding:** A diode was used to remove the negative half cycles. We observed an offset of about 500 mV at the diode output, which was used as $V_{ref}$ for a comparator to map the waveform to logic levels (approximately +5 V for '1' and 0 V for '0').
- **Reference voltage generation:** Initially, a resistive divider was used for $V_{ref}$, but it was difficult to set precisely to 500 mV. We therefore switched to a potentiometer, which allowed smooth and accurate adjustment of the threshold.
- **Recovered output:** The output of the comparator provided the final demodulated (recovered) binary sequence.